

INFLUENCE OF DIFFERENT RELATIVE INTENSITIES ON POWER OUTPUT DURING THE HANG POWER CLEAN: IDENTIFICATION OF THE OPTIMAL LOAD

NAOKI KAWAMORI,¹ AARON J. CRUM,¹ PETER A. BLUMERT,¹ JUSTIN R. KULIK,¹
JAMES T. CHILDERS,¹ JULIE A. WOOD,¹ MICHAEL H. STONE,² AND G. GREGORY HAFF³

¹Human Performance Laboratory, Midwestern State University, Wichita Falls, Texas 76308; ²Coaching and Sport Science Division, United States Olympic Committee, Colorado Springs, Colorado 80909; ³Division of Exercise Physiology, West Virginia University School of Medicine, Morgantown, West Virginia 26506.

ABSTRACT. Kawamori, N., A.J. Crum, P.A. Blumert, J.R. Kulik, J.T. Childers, J.A. Wood, M.H. Stone, and G.G. Haff. Influence of different relative intensities on power output during the hang power clean: Identification of the optimal load. *J. Strength Cond. Res.* 19(3):698–708, 2005.—The influence of different relative intensities on power output was investigated in the present study in order to identify the optimal load that maximizes power output during the hang power clean. Fifteen men (age: 22.1 ± 2.0 years, height: 180.1 ± 6.3 cm, and body mass: 89.4 ± 14.7 kg) performed the hang power cleans on a forceplate at 30–90% of one repetition maximum (1RM). Peak power was maximized at 70% 1RM, which was, however, not significantly different from peak power at 50, 60, 80, and 90% 1RM. Average power also was maximized at 70% 1RM, which was not significantly different from average power at 40, 50, 60, 80, and 90% 1RM. It was concluded that (a) the relative intensity had a significant influence on power output, and (b) power output can be maximized at a submaximal load during the hang power clean.

KEY WORDS. force-time curve, rate of force development, vertical jump, resistance training, ground reaction force

INTRODUCTION

Muscular power is considered one of the main determinants of dynamic athletic performance, especially in sporting events that require high force generation in a short amount of time (39). Therefore, training aimed at power improvement is essential for athletes. Power is defined as the rate of doing work, and can be expressed as the product of force and velocity, which have an inverse relationship in concentric muscle actions: As the velocity of movement increases, the force that a muscle can produce decreases. Therefore, the highest power during a movement (peak power) is achieved at a compromised level of force and velocity. It has been reported that the highest power output occurs at approximately 30% of maximum isometric force during single-joint movement (29).

Of particular importance for the athlete and coach is the optimum load that maximizes peak power during commonly used, dynamic resistance-training exercises. Interestingly, the results of recent studies indicate that a relatively wide range of loads can maximize power output during different types of resistance-training movements (4, 5, 10, 26, 42, 48, 50). For example, Stone et al. (48) found that in a heterogeneously trained group, the highest power output occurs at 10% of one repetition maximum (1RM) during both countermovement and static

squat jumps, whereas Cronin et al. (10) found that the highest power output occurred at 50–70% 1RM during the bench press and bench throw. Therefore, the optimal load that maximizes power output may differ between selected exercises. If this is the case, identifying the optimal load for each exercise is important, because training with the optimal load is suggested to be the most effective method for improving maximal power and a variety of dynamic athletic performance (29, 53). Another factor that influences the optimal load deals with the trained state: some data indicate that strength training may shift the percentage of maximum strength at which the highest power output occurs (3, 48).

Weightlifting movements (WLM) and their derivatives generally are considered to be a superior resistance-training method for muscular power development and athletic performance enhancement (16, 20, 46). Weightlifting movements and their derivatives are known to produce some of the highest average human power outputs of all the resistance-training exercises (20, 46). For example, at maximum load, a 100-kg male weightlifter can produce an absolute power output of 3,000 W in the snatch, as compared with 1,100 W in the squat (46). In addition to a high power-producing capability, high mechanical specificity makes WLM and their derivatives effective training exercises for athletes who need explosive strength and power to be successful in their sports (15). Mechanical specificity refers to the kinetic and kinematic similarity of training exercises to the actual athletic performance (49). It is believed that the more specific a training exercise is to the actual athletic performance, the greater the transfer of training effect (1, 49). Generally, WLM and their derivatives are considered highly specific to actual sport performance, because they involve large muscle mass, multi-joint movements, and fast movement velocity (1, 49). Additionally, biomechanical comparisons of WLM and sport-related performance measures (e.g., vertical jump) indicate a high degree of mechanical similarity between activities (8, 17).

Regarding the influence of relative intensity on power output during WLM and their derivatives, Garhammer (16) indicates that power output increases as the weight lifted decreases from the maximal load. It was further suggested that relatively high loads of 75–85% 1RM will maximize power output during WLM and their derivatives. However, traditional nonballistic exercises such as the squat and bench press generally maximize power out-

put at 30–60% 1RM (26, 42). Part of the reason for this difference is that WLM are ballistic in nature and are velocity-dependent (18).

Haff et al. (18) required subjects to perform the clean pull from midhigh at 80, 90, and 100% 1RM of the power clean. They found a general trend of increasing power output as the relative intensity decreased from 100% to 80%, which agrees with Garhammer's findings (16). Therefore, it seems feasible that the highest power output is achieved at submaximal loads, rather than at near-maximal loads during WLM and their derivatives. However, little data exist in the scientific literature investigating the exact relative intensity that maximizes power output during WLM and their derivatives (18, 21).

Therefore, the purpose of this investigation was to examine (a) the influence of different relative intensities on power output and to identify the optimal load for the highest peak and average power outputs during the hang power clean, one derivative of WLM; (b) to investigate the relationship between power output during the hang power clean and dynamic athletic performance (vertical jump displacement) (49); and (c) to investigate the influence of the hang power clean 1RM on the optimal load, to determine if individuals with a higher 1RM achieve the higher power output at different relative intensities than individuals with a lower 1RM, as suggested by Baker (3) and Stone et al. (48).

METHODS

Experimental Approach to the Problem

The present study was designed to investigate the influence of different relative intensities on power output during the hang power clean. Subjects participated in 1 familiarization session and 2 testing sessions separated by at least 2 days. The first testing session was used to determine biometric data, static (SJ) and countermovement vertical jump (CMJ) performance, and subjects' 1RM in the hang power clean. The second testing session was used to record the ground reaction force during the hang power clean at different relative intensities from 30–90% 1RM in 10% increments. Subjects were asked to refrain from physical activity that might deteriorate testing performance for 48 hours prior to each testing session.

Subjects

Fifteen men who were familiar with the hang power clean volunteered to be subjects for the present investigation. Subjects included 8 National Collegiate Athletic Association Division II football players, 3 weightlifters, a rugby player, a bobsledder, a basketball player, and a recreationally trained man. Subjects were required to have a minimum of 6 months' experience performing the hang power clean prior to participation in this study. Subjects' biometric data are provided in Table 1. Prior to participating in this investigation, all subjects read and signed an informed consent and completed a health history questionnaire, in accordance with guidelines set forth by the Human Subjects Review Committee at Midwestern State University.

Biometric Data

Subjects had their height, body mass, and body composition measured. Height was measured using a stadiometer to the nearest 0.1 cm, and body mass was determined

TABLE 1. Subject characteristics ($n = 15$).

Variable	Mean $\pm$ SD
Age (y)	22.1 $\pm$ 2.0
Height (cm)	180.1 $\pm$ 6.3
Body mass (kg)	89.4 $\pm$ 14.7
Body fat (%)	8.5 $\pm$ 3.7
1RM hang power clean (kg)*	107.0 $\pm$ 18.8
1RM hang power clean (kg·BdM ⁻¹)	1.20 $\pm$ 0.15

* 1RM = 1 repetition maximum; BdM = body mass.

using an electronic scale to the nearest 0.1 kg. A 7-site skinfold measurement was used to assess body composition. Skinfolds were measured using Harpenden Skinfold Calipers (Baty International, England) on the right side of the body, 3 times at each site, with the mean value recorded. The 7 sites used were the triceps, subscapular, chest, suprailiac, midaxillary, abdomen, and thigh. Body density was calculated (28), and then the Siri equation was used to estimate percentage of body fat for each subject (44).

Vertical Jump Testing

For the vertical jump tests, subjects performed a dynamic warm-up, which was similar to procedures reported previously by Kirksey et al. (31). Subjects then performed the CMJ and SJ on a 61 $\times$ 121.9-cm AMTI forceplate (Advanced Mechanical Technologies, Newton, MA). Each subject began the CMJ in the standing position, dropped into the squat position, and immediately jumped vertically as high as possible. The depth of knee flexion during the CMJ was self-determined. Hands were kept on the hips during the CMJ. Each subject performed 2 CMJs using maximal effort, with a 1-minute rest between trials. The average value of the 2 trials was used later in the analysis process. The same protocol was repeated for the SJ, except that each subject began the SJ from a squat position with a knee angle of 90° (determined by tester observation). This position was held for 3 seconds before initiating a maximum upward effort. No countermovement was allowed during the SJ.

One Repetition Maximum Testing

Subjects were tested for their 1RM hang power clean using previously established methods by Stone and O'Bryant (47). The hang power clean was initiated from the standing position with a subject holding the bar. The subject began the hang power clean by lowering the bar to midhigh above the knee, and then lifted the bar explosively upward with maximum effort to receive it on the shoulders in a quarter-squat position. Any hang power clean received in a squat position with the upper thigh below parallel to the floor was ruled unsuccessful.

Power Testing

Power testing began with an adequate warm-up of a dynamic nature. Next, subjects performed single maximal-effort hang power cleans on a forceplate at loads of 30, 40, 50, 60, 70, 80, and 90% 1RM in that order, with 2 attempts made at each load. The hang power cleans were performed in the same manner as used in the 1RM test. Subjects were instructed to lift the barbell as fast and hard as possible in an attempt to maximize power output (6, 18). The average value of the 2 trials at each load was

used later in the analysis process. Adequate recovery of a minimum of 3 minutes was provided between trials to assure maximum effort during subsequent trials.

Instruments

A 61 × 121.9-cm AMTI forceplate (Advanced Mechanical Technologies, Newton, MA), which sampled at 600 Hz, was used to record the ground reaction force during vertical jumps and the hang power cleans. The forceplate was calibrated prior to each testing session with standard techniques.

Data Analyses

From the vertical component of the recorded ground reaction force during the hang power cleans, peak and average power, peak and average velocity of the center of mass of the lifter-plus-bar system, peak rate of force development (PRFD), time to peak force, time to PRFD, and the duration of the concentric phase of the hang power clean were calculated using methods previously described in the scientific literature (18, 22, 23). Specifically, power output during the hang power clean was calculated as follows: Power (W) = vertical ground reaction force (N) × vertical velocity of the center of mass of the lifter-plus-bar system (m·s⁻¹).

Vertical velocity of the center of mass of the lifter-plus-bar system was calculated using the principle that the impulse is equal to the change in momentum. Thus, the change in vertical velocity of the center of mass of the system over a sampling interval was the net vertical ground reaction force multiplied by the intersample time period divided by the mass of the system. The net vertical ground reaction force was taken as the vertical force platform reading (i.e., vertical ground reaction force) minus the weight of the lifter-plus-bar system. The absolute vertical velocity of the center of mass of the system at the end of each sampling interval was then determined by adding the change in vertical velocity over a sampling interval to the preinterval absolute velocity, which was zero at the start of the movement. After the variables of force, velocity, and power were established for the entire movement, each variable's peak and average values were determined for the concentric phase of the hang power clean. Initiation of the concentric phase was determined as the time when the vertical velocity of the center of mass of the system changed from negative to positive, and the end of the concentric phase was determined as the time when the vertical component of the ground reaction force became equal to the weight of the lifter-plus-bar system immediately before the subject caught the bar on the shoulders (12). Test-retest reliabilities for peak power, average power, peak ground reaction force, average ground reaction force, peak velocity of the center of mass of the lifter-plus-bar system, average velocity of the center of mass of the lifter-plus-bar system, and the duration of the concentric phase were intra-class correlation (ICC) = 0.98, ICC = 0.98, ICC = 0.99, ICC = 0.99, ICC = 0.94, ICC = 0.97, and ICC = 0.94, respectively. In addition, rate of force development (RFD) was calculated as the first derivative of the recorded force-time curve for the entire movement (35), and then the peak value was determined. Time to peak force and to PRFD also was calculated with the initiation of calculation determined as the point where the minimum force value was achieved during the movement (52). Test-retest reliabilities for

PRFD, time to peak force, and time to PRFD were ICC = 0.95, ICC = 0.96, and ICC = 0.98, respectively.

From the vertical component of the recorded ground reaction force during the vertical jump tests, vertical jump displacement was calculated from the measured flight times using the following equation (7):

$$\text{Vertical jump displacement} = \frac{g \times \text{flight time} \times \text{flight time}}{8},$$

where *g* is the acceleration due to gravity. Because the use of this equation assumes the same body position of the subject at takeoff and landing, subjects were instructed to land on the ground with their knees extended and to touch the forceplate first with their toes after the jump (33). Test-retest reliability for vertical jump displacement was ICC = 0.99 (CMJ) and ICC = 0.94 (SJ). In addition, peak power, peak force, and PRFD also were calculated from the vertical component of the recorded ground reaction force during the vertical jump tests using the same methods described above for the hang power clean analysis. Test-retest reliabilities for vertical jump peak power, peak force, and PRFD were ICC = 0.98 (CMJ) and ICC = 0.98 (SJ), ICC = 0.99 (CMJ) and ICC = 0.98 (SJ), and ICC = 0.95 (CMJ) and ICC = 0.96 (SJ), respectively.

Statistical Analyses

A repeated measures 1-way analysis of variance (ANOVA) was used to determine significant differences between different relative intensities for peak power, average power, peak ground reaction force, average ground reaction force, peak velocity of the center of mass of the lifter-plus-bar system, average velocity of the center of mass of the lifter-plus-bar system, PRFD, time to peak force, time to PRFD, and the duration of the concentric phase of the hang power clean. When significant *F* values were found (*p* ≤ 0.05), paired comparisons were used in conjunction with the Holm's Bonferroni method for controlling type I error (24) to determine the significant differences. Pearson's product moment correlation coefficients were used to determine the relationship between variables during the hang power clean and the vertical jump. The relative strength of the correlations were based on the work of Cohen (9) and Hopkins (25). In order to examine the effect of 1RM hang power clean on optimal load, subjects were divided into subgroups of 7 strong (≥110 kg) and 8 weak (<110 kg) subjects, based on 1RM hang power clean performance. Data from the strong and weak groups were compared using Student's *t*-tests in order to determine if significant differences existed in 1RM hang power clean, peak power, and average power between these two subgroups. All data were reported as mean ± standard deviation (*SD*). The level of significance was set at *p* ≤ 0.05 in this study. All statistics were performed using SPSS 12.1 (SPSS Inc., Chicago, IL).

RESULTS

Figure 1 depicts the force-time curve for a representative subject performing the hang power clean. The performance characteristics of the CMJ and SJ are presented in Table 2.

Influence of Relative Intensity During the Hang Power Clean

The statistical results generated by the ANOVA confirmed that the relative intensity (% 1RM) had a signifi-

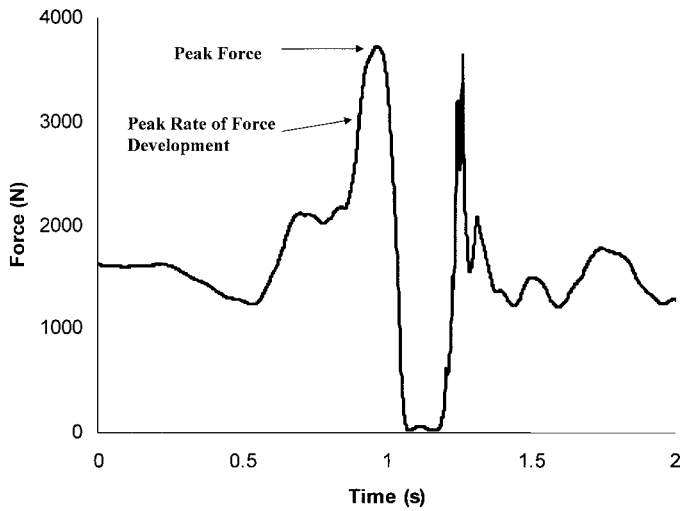


FIGURE 1. Force-time curve for a representative subject performing the hang power clean.

TABLE 2. Performance characteristics of vertical jump tests ($n = 15$).

Variable	Countermovement vertical jump	Static vertical jump
	Mean $\pm$ SD	Mean $\pm$ SD
Displacement (m)	0.42 $\pm$ 0.09	0.37 $\pm$ 0.07
Peak power (W)	5,380.51 $\pm$ 727.19	5,219.40 $\pm$ 680.47
Peak power (W·kg ⁻¹)*	60.9 $\pm$ 8.2	59.05 $\pm$ 7.88
Peak force (N)	2,283.48 $\pm$ 344.41	2,229.47 $\pm$ 318.48
Peak force (N·kg ⁻¹)*	25.7 $\pm$ 2.3	25.12 $\pm$ 2.92
Peak RFD (N·s ⁻¹)†	14,010.51 $\pm$ 6,318.52	12,339.63 $\pm$ 5,150.13
Peak RFD (N·s ⁻¹ ·kg ⁻¹)*	157.8 $\pm$ 75.6	138.3 $\pm$ 56.0

* = values relative to body mass.

† RFD = rate of force development.

cant effect on peak power ($p < 0.001$, $\eta^2 = 0.97$, $1 - \beta = 1.00$). Follow-up paired comparisons revealed that peak power rates at 50, 60, 70, 80, and 90% 1RM were significantly different from peak power at 30% 1RM, and peak power rates at 50, 60, 70, and 80% 1RM were significantly different from peak power at 40% 1RM (Figure 2). Additionally, relative intensity had a significant effect on average power ($p < 0.01$, $\eta^2 = 0.85$, $1 - \beta = 0.98$). Follow-up tests found that average power rates at 40, 50, 60, 70,

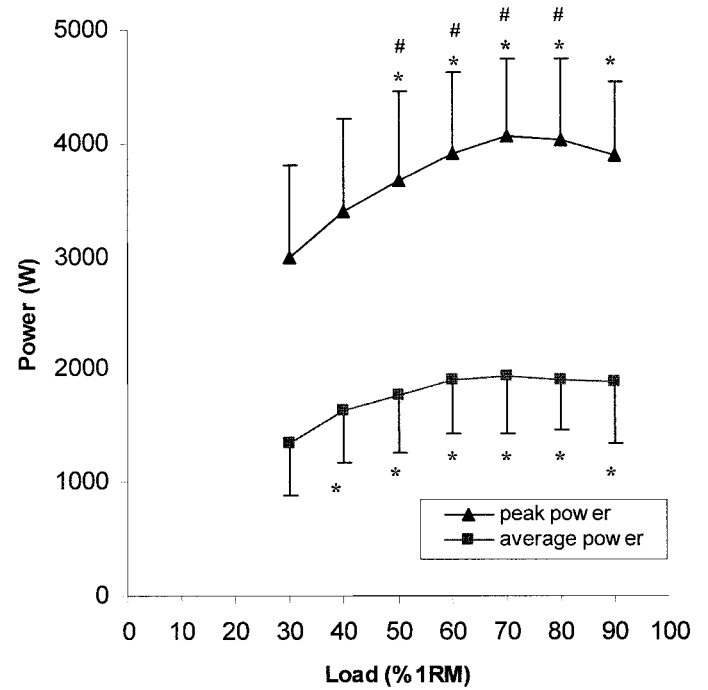


FIGURE 2. Peak and average power at loads of 30–90% of one repetition maximum (1RM) during the hang power clean. * = significantly different from 30%; # = significantly different from 40%.

80, and 90% 1RM were significantly different from average power rates at 30% 1RM (Figure 2). When converting peak power measures to relative values (peak power per body mass), there was a significant difference between loads ($p < 0.001$, $\eta^2 = 0.96$, $1 - \beta = 1.00$). Follow-up tests indicated that peak power per body mass at 50, 60, 70, 80, and 90% 1RM was significantly different from peak power per body mass at 30% 1RM, and peak power per body mass at 50, 60, 70, and 80% 1RM was significantly different from peak power per body mass at 40% 1RM (Table 3). Additionally, when converting average power measures to relative values (average power per body mass), there was a significant difference between loads ($p < 0.01$, $\eta^2 = 0.84$, $1 - \beta = 0.98$). Follow-up paired comparisons revealed that average power per body mass at 40, 50, 60, 70, 80, and 90% 1RM was significantly differ-

TABLE 3. Relative values of performance characteristics of the hang power clean trials ($n = 15$).

Load (% 1RM)	Peak power (W·kg ⁻¹)	Average power (W·kg ⁻¹)	Peak force (N·kg ⁻¹)§	Average force (N·kg ⁻¹)	Peak RFD* (N·s ⁻¹ ·kg ⁻¹)
	Mean $\pm$ SD	Mean $\pm$ SD	Mean $\pm$ SD	Mean $\pm$ SD	Mean $\pm$ SD
30	33.44 $\pm$ 7.53	14.87 $\pm$ 4.09	29.89 $\pm$ 3.70	22.33 $\pm$ 3.04	219.60 $\pm$ 95.90
40	38.06 $\pm$ 7.89	18.31 $\pm$ 4.75†	31.92 $\pm$ 4.06	24.97 $\pm$ 3.16	219.29 $\pm$ 96.26
50	41.04 $\pm$ 7.20†‡	19.80 $\pm$ 4.89†	34.55 $\pm$ 4.00	26.80 $\pm$ 3.36	232.46 $\pm$ 96.35
60	43.87 $\pm$ 6.50†‡	21.34 $\pm$ 4.75†	36.37 $\pm$ 3.79	28.50 $\pm$ 3.43	234.50 $\pm$ 94.60
70	45.57 $\pm$ 5.20†‡	21.62 $\pm$ 4.63†	37.31 $\pm$ 3.64	29.35 $\pm$ 3.55	215.72 $\pm$ 77.19
80	45.24 $\pm$ 6.42†‡	21.32 $\pm$ 4.07†	37.93 $\pm$ 3.58	30.12 $\pm$ 3.49	225.26 $\pm$ 79.96
90	43.76 $\pm$ 5.23†‡	21.16 $\pm$ 5.11†	38.49 $\pm$ 3.58	30.99 $\pm$ 3.76	231.64 $\pm$ 85.53

* RFD = rate of force development; 1RM = 1 repetition maximum.

† = significantly different from 30%.

‡ = significantly different from 40%.

§ = peak forces at different relative intensities are significantly different except for between 70 and 80% 1RM and between 80 and 90% 1RM.

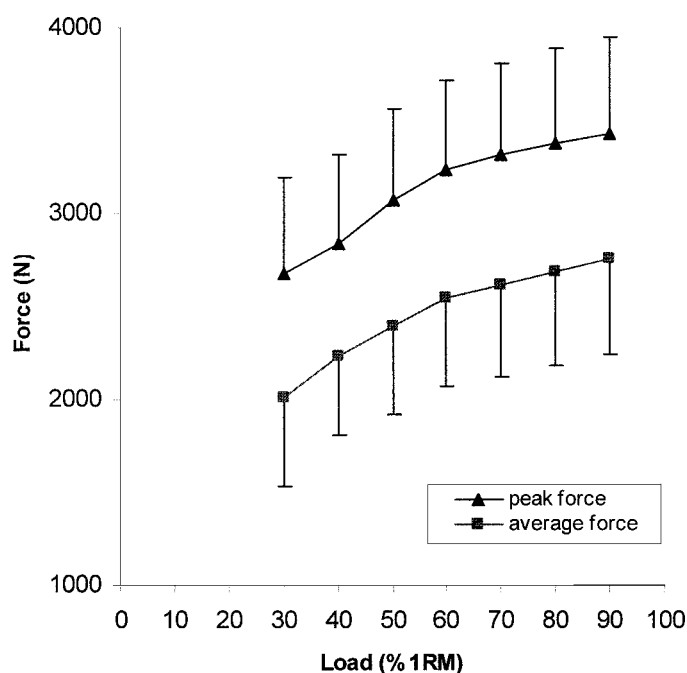


FIGURE 3. Peak and average ground reaction force at loads of 30–90% 1RM during the hang power clean. All values are significantly different from each other except for peak power between 70 and 80% 1RM and between 80 and 90% 1RM.

ent from average power per body mass at 30% 1RM (Table 3).

The results of the statistical analyses indicated that the relative intensity significantly influenced peak ground reaction force ($p < 0.001$, $\eta^2 = 0.95$, $1 - \beta = 1.00$). Follow-up paired comparisons showed that peak force significantly increased as the load increased, except for between 70 and 80% 1RM and between 80 and 90% 1RM (Figure 3). The ANOVA ($p < 0.001$, $\eta^2 = 0.94$, $1 - \beta = 1.00$) and follow-up tests showed that average force also increased significantly as the load increased (Figure 3). When converting peak force measures to relative values (peak force per body mass), there was a significant difference between loads ($p < 0.001$, $\eta^2 = 0.93$, $1 - \beta = 1.00$). Follow-up paired comparisons indicated that peak force per body mass increased as the load increased, except for between 70 and 80% 1RM and between 80 and 90% 1RM (Table 3). In addition, when converting average force measures to relative values (average force per body mass), there was a significant difference between loads ($p < 0.001$, $\eta^2 = 0.92$, $1 - \beta = 1.00$). Follow-up tests revealed that average force per body mass increased as the load increased (Table 3).

Significant differences in peak velocity ($p < 0.01$, $\eta^2 = 0.88$, $1 - \beta = 1.00$) and average velocity ($p < 0.05$, $\eta^2 = 0.77$, $1 - \beta = 0.87$) of the center of mass of the lifter-plus-bar system between different relative intensities were indicated by an ANOVA. Follow-up tests revealed that peak velocity at 60 and 70% 1RM was significantly different from peak velocity at 90% 1RM (Figure 4). In addition, average velocity at 50 and 60% 1RM was significantly different from average velocity at 30% 1RM (Figure 4).

When comparing the PRFD achieved during the hang power clean, there was no statistically significant difference between the different relative intensities ($p > 0.05$,

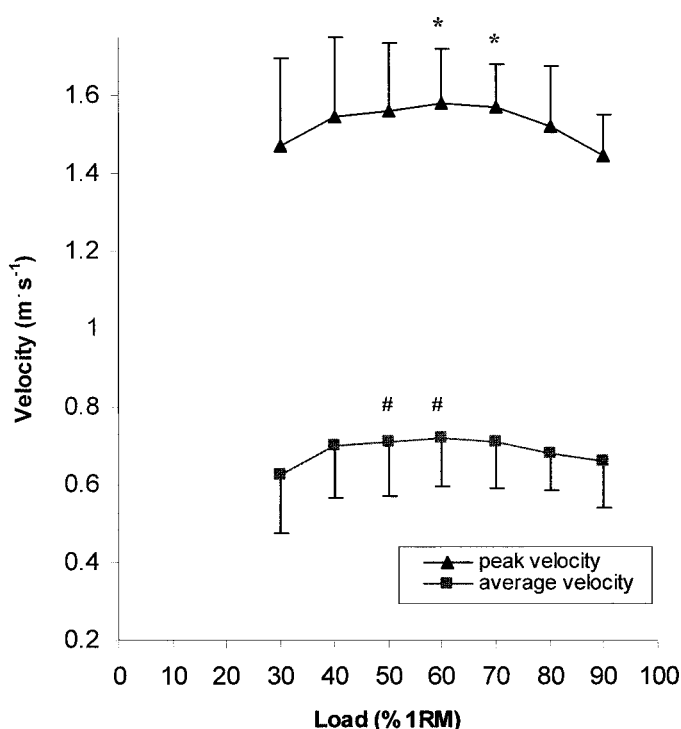


FIGURE 4. Peak and average velocity of the center of mass of the lifter-plus-bar system. * = significantly different 90%; # = significantly different from 30%.

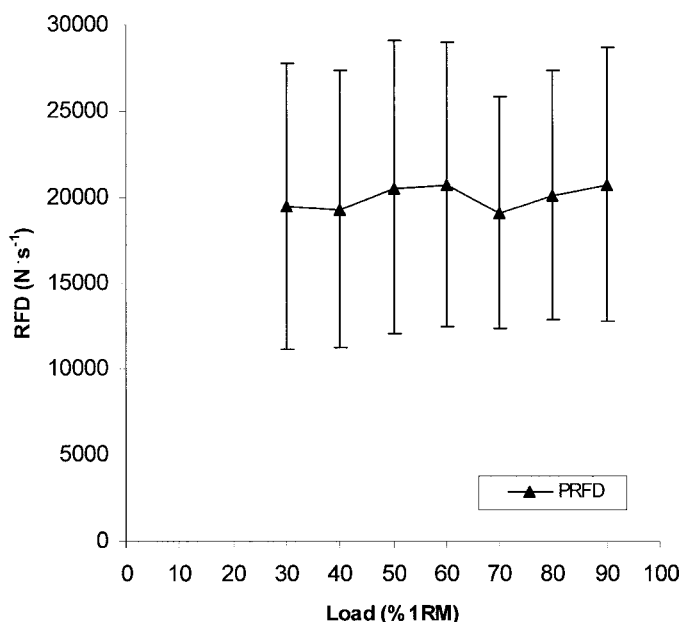


FIGURE 5. Peak rate of force development (RFD) at loads of 30–90% 1RM during the hang power clean.

$\eta^2 = 0.38$, $1 - \beta = 0.21$) (Figure 5). Additionally, when converting the PRFD measures to relative values (PRFD per body mass), there was no significant difference between the intensities ($p > 0.05$, $\eta^2 = 0.36$, $1 - \beta = 0.20$) (Table 3).

Additionally, there was no significant difference between the different relative intensities during the hang power clean in time to peak force ($p > 0.05$, $\eta^2 = 0.31$, $1 - \beta = 0.21$) (Table 3).

TABLE 4. Time-related performance characteristics of the hang power clean trials ($n = 15$).

Load (% 1RM)	Time to peak force (ms)	Time to peak RFD (ms)*	Duration of concentric phase (ms)
	Mean $\pm$ SD	Mean $\pm$ SD	Mean $\pm$ SD
30	535.77 $\pm$ 228.92	473.63 $\pm$ 247.92	265.83 $\pm$ 175.39
40	543.60 $\pm$ 240.99	444.17 $\pm$ 279.79	233.60 $\pm$ 74.27
50	537.07 $\pm$ 208.71	464.53 $\pm$ 219.34	239.90 $\pm$ 55.36
60	540.10 $\pm$ 228.29	453.43 $\pm$ 239.36	245.30 $\pm$ 53.60
70	577.50 $\pm$ 254.18	456.90 $\pm$ 268.98	268.90 $\pm$ 58.08†
80	591.50 $\pm$ 283.10	477.57 $\pm$ 306.97	290.47 $\pm$ 82.76†‡§
90	613.10 $\pm$ 310.99	437.13 $\pm$ 359.98	303.83 $\pm$ 84.31†‡§

* RFD = rate of force development; 1RM = 1 repetition maximum.

† = significantly different from 40%.

‡ = significantly different from 50%.

§ = significantly different from 60%.

TABLE 5. Correlations (r) between 1 repetition maximum (1RM) hang power clean and vertical jump performance variables ($n = 15$).

Hang power clean	Countermovement vertical jump				Static vertical jump			
	Disp.*	PP	PF	PRFD	Disp.	PP	PF	PRFD
1RM (kg)	0.13	0.68‡	0.87‡	0.53†	0.09	0.71‡	0.82†	0.68‡
1RM (kg·BdM ⁻¹)	0.56†	0.45	0.23	0.56†	0.47	0.46	0.39	0.64‡

* Disp. = displacement; PP = peak power; PF = peak force; PRFD = peak rate of force development; BdM = body mass.

† = significant at $p < 0.05$.

‡ = significant at $p < 0.01$.

– $\beta = 0.16$) or in time to PRFD ($p > 0.05$, $\eta^2 = 0.18$, $1 - \beta = 0.10$; Table 4). When comparing the duration of the concentric phase during the hang power clean, there was a significant difference between the different relative intensities ($p < 0.05$, $\eta^2 = 0.75$, $1 - \beta = 0.83$). Follow-up paired comparisons indicated that the duration of the concentric phase at 80 and 90% 1RM were significantly different from the duration of the concentric phase at 40, 50, and 60% 1RM. Additionally, the duration of the concentric phase at 70% 1RM was significantly different from the duration of the concentric phase at 50% 1RM (Table 4).

Correlation Analyses

One repetition maximum of the hang power clean had no significant relationship with CMJ and SJ displacement (Table 5). When converted to values relative to body mass, a significant correlation was found between 1RM hang power clean per body mass and CMJ displacement ($r = 0.56$, $r^2 = 0.31$, $p < 0.05$), but not between 1RM hang power clean per body mass and SJ displacement.

Neither peak nor average power during the hang power clean was significantly correlated with CMJ and SJ displacement (Table 6). When power outputs were converted to values relative to body mass, peak power during the hang power clean at 70% ($r = 0.68$, $r^2 = 0.46$, $p < 0.05$), 80% ($r = 0.58$, $r^2 = 0.34$, $p < 0.05$), and 90% 1RM ($r = 0.72$, $r^2 = 0.52$, $p < 0.05$) was significantly correlated with CMJ displacement, and peak power during the hang power clean at 70% ($r = 0.53$, $r^2 = 0.28$, $p < 0.05$) and 90% ($r = 0.52$, $r^2 = 0.27$, $p < 0.05$) was significantly correlated with SJ displacement (Table 7). In addition, peak force and PRFD showed no significant correlations with CMJ or SJ displacement (Table 8).

Influence of 1RM on the Optimal Load

Table 9 shows the comparisons of hang power clean 1RM and peak and average power during the hang power clean between 7 strong (1RM ≥ 110 kg) and 8 weak (1RM < 110 kg) subjects. Subjects in the strong subgroup had significantly higher hang power clean 1RMs in both absolute and relative values, as compared with subjects in the weak subgroup. Strong subjects achieved the highest peak power at 70% 1RM during the hang power clean, whereas weak subjects maximized peak power at 80% 1RM. In addition, strong subjects achieved the highest average power at 70% 1RM during the hang power clean, whereas weak subjects maximized average power at 60% 1RM.

DISCUSSION

The primary finding of this investigation is that peak and average power during the hang power clean were maximized at 70% 1RM. However, peak power at 70% 1RM was not significantly different from peak power at 50, 60, 80, and 90% 1RM, and average power at 70% 1RM was not significantly different from average power at 40, 50, 60, 80 and 90% 1RM. The relative intensity of 70% 1RM, as the optimal load for the highest power output, is considered higher than traditionally reported values of 15–45% of maximum isometric force or 1RM (13, 29, 40). Several other investigators have reported comparably high loads of 55–70% 1RM as the optimal load to maximize power output during various resistance-training movements (4, 5, 27, 32, 50). Possible factors that might explain these discrepancies include single vs. multiple joint movements, the type of muscle action (e.g., concentric, plyometric), the mode of exercise, the strength level of

TABLE 6. Correlations (r) between power output during the hang power clean and vertical jump performance variables ($n = 15$).

Hang power clean	Counter movement vertical jump				Static vertical jump			
	Disp.*	PP	PF	PRFD	Disp.	PP	PF	PRFD
Peak power (W)								
At 30%	0.00	0.55†	0.66‡	0.24	0.13	0.47	0.69‡	0.39
At 40%	0.09	0.52†	0.76‡	0.55†	0.08	0.49	0.70‡	0.46
At 50%	-0.04	0.40	0.70‡	0.43	-0.03	0.42	0.76‡	0.46
At 60%	0.16	0.59†	0.69‡	0.38	0.13	0.61†	0.85‡	0.52†
At 70%	0.22	0.72‡	0.83‡	0.46	0.12	0.70‡	0.77‡	0.42
At 80%	0.26	0.68‡	0.59†	0.21	0.21	0.63†	0.65‡	0.33
At 90%	0.28	0.71‡	0.72‡	0.41	0.15	0.72‡	0.74‡	0.42
Average power (W)								
At 30%	-0.08	0.30	0.72‡	0.46	-0.13	0.47	0.80‡	0.66‡
At 40%	-0.11	0.27	0.66‡	0.47	-0.17	0.35	0.71‡	0.57†
At 50%	-0.17	0.20	0.59†	0.35	-0.29	0.34	0.67‡	0.47
At 60%	-0.01	0.34	0.61†	0.37	-0.22	0.47	0.72‡	0.54†
At 70%	0.04	0.47	0.70‡	0.38	-0.18	0.58†	0.68‡	0.43
At 80%	0.12	0.55†	0.66‡	0.28	-0.08	0.66‡	0.70‡	0.39
At 90%	0.02	0.42	0.58†	0.26	-0.12	0.51	0.60*	0.37

* Disp. = displacement; PP = peak power; PF = peak force; PRFD = peak rate of force development.

† = significant at $p < 0.05$.‡ = significant at $p < 0.01$.

individual subjects, the training status of individuals within the yearly training cycle, and how power output was calculated.

Each resistance exercise involves different muscle groups, different ranges of motion, different techniques, and different types of muscle action. Such differences in the mode of exercise might lead to different optimal loads for the highest power output (30). Indeed, differences in optimal loads and power-load curves have been reported between different resistance-training exercises (26, 42). In addition, it has been suggested that stronger individuals maximize power output at different relative intensities than less strong individuals do (3, 48). This indicates that prolonged training can shift the optimal load either upward or downward, based on the type of resistance-training programs and sporting events in which athletes have been involved (3, 26, 48). The shift of the optimal load might be apparent even after a relatively short period of resistance training. For example, Duchateau and Hainaut (11) found that 3 months of isometric training shifted the optimal load toward heavier loads. Baker (2) also suggested that the optimal load fluctuates according to the training state of athletes within a yearly cycle.

Different methods of calculating power output can change power output values and might affect the load that maximizes power output. Sleivert and Taingahue (45) suggest that including or excluding body mass in power calculations, and using the velocity of the bar or the velocity of the center of mass of the lifter-plus-bar system for power calculation, can affect the relative intensity at which the highest power occurs, especially when exercises are performed in the vertical plane. Therefore, one needs to be aware of the methods used to calculate power outputs when looking at the results of different studies that investigated the influence of relative intensities on power outputs during various resistance-training exercises.

Identifying the optimal load for the highest power output during resistance-training movements has practical significance for athletes, because resistance training at the optimal load appears to be the most effective method for improving maximal muscular power and a wide range of dynamic athletic performance variables (29, 53). Kaneko et al. (29) investigated the effects of different training loads on mechanical power output. Subjects trained elbow flexors either at 0, 30, 60, or 100% of maximum isometric force with maximum effort. They found that the maximal mechanical power output, which appeared at 30% of maximum isometric force, was improved most by training at 30% of maximum isometric force. Additionally, Wilson et al. (53) compared the effects of traditional weight training (i.e., heavy squat lifts), plyometric training (i.e., depth jumps), and explosive weight training at the load that maximized mechanical power output (i.e., weighted squat jumps) on a series of dynamic athletic performance variables including the 30-m sprint, vertical jump, and a maximal cycle test. It was shown that the experimental group that used weighted squat jumps as a training modality achieved the best overall results in enhancing dynamic athletic performance variables. Based on the finding of these 2 studies, it can be argued that athletes should use a training load that maximizes power output in order to improve muscular power and a wide variety of dynamic athletic performance variables.

TABLE 7. Correlations (r) between power output during the hang power clean in relative values and vertical jump performance variables ($n = 15$).

Hang power clean	Countermovement vertical jump				Static vertical jump			
	Disp.*	PP	PF	PRFD	Disp.	PP	PF	PRFD
Peak power (W·kg ⁻¹)								
At 30%	0.23	0.43	0.30	0.20	0.38	0.34	0.49	0.32
At 40%	0.37	0.35	0.32	0.56*	0.35	0.31	0.42	0.40
At 50%	0.23	0.17	0.18	0.40	0.24	0.18	0.44	0.37
At 60%	0.48	0.34	0.08	0.30	0.43	0.35	0.45	0.37
At 70%	0.68‡	0.52†	0.17	0.43	0.53†	0.47	0.33	0.26
At 80%	0.58†	0.43	-0.05	0.06	0.50	0.36	0.19	0.13
At 90%	0.72‡	0.45	0.01	0.31	0.52†	0.44	0.23	0.23
Average power (W·kg ⁻¹)								
At 30%	0.17	0.22	0.43	0.51	0.09	0.43	0.71†	0.72‡
At 40%	0.12	0.10	0.26	0.47	0.03	0.18	0.47	0.53†
At 50%	0.06	0.03	0.18	0.34	-0.11	0.18	0.45	0.44
At 60%	0.23	0.12	0.13	0.32	-0.35	0.25	0.42	0.44
At 70%	0.33	0.33	0.27	0.36	0.03	0.45	0.43	0.37
At 80%	0.42	0.39	0.17	0.20	0.14	0.50	0.41	0.29
At 90%	0.25	0.23	0.12	0.18	0.07	0.32	0.30	0.27

* Disp. = displacement; PP = peak power; PF = peak force; PRFD = peak rate of force development.

† = significant at $p < 0.05$.

‡ = significant at $p < 0.01$.

TABLE 8. Correlations (r) between force-time curve variables during the hang power clean and vertical jump performance variables ($n = 15$).

Hang power clean	Countermovement vertical jump				Static vertical jump			
	Disp.*	PP	PF	PRFD	Disp.	PP	PF	PRFD
Peak force (N)								
At 30%	0.00	0.58†	0.83‡	0.36	0.00	0.61†	0.81‡	0.45
At 40%	0.03	0.52†	0.76‡	0.39	0.01	0.57†	0.80‡	0.42
At 50%	-0.09	0.40	0.64†	0.26	-0.18	0.49	0.74‡	0.29
At 60%	-0.05	0.46	0.65‡	0.21	-0.17	0.55†	0.79‡	0.32
At 70%	-0.01	0.54†	0.68‡	0.21	-0.12	0.61†	0.77‡	0.30
At 80%	-0.06	0.50	0.67‡	0.15	-0.16	0.57†	0.80‡	0.34
At 90%	-0.07	0.50	0.67‡	0.16	-0.18	0.56†	0.78‡	0.35
PRFD (N·s ⁻¹)								
At 30%	0.21	0.31	0.19	0.04	0.16	0.30	0.30	-0.01
At 40%	0.33	0.36	0.12	0.07	0.39	0.34	0.32	0.08
At 50%	0.14	0.22	0.02	-0.07	0.03	0.23	0.24	-0.21
At 60%	0.15	0.17	-0.08	-0.19	-0.04	0.22	0.30	-0.12
At 70%	0.00	0.00	-0.17	-0.33	-0.22	0.08	0.21	-0.17
At 80%	-0.20	-0.12	-0.03	-0.26	0.42	-0.01	0.37	0.06
At 90%	-0.28	-0.02	0.10	-0.22	-0.45	-0.04	0.34	0.09

* Disp. = displacement; PP = peak power; PF = peak force; PRFD = peak rate of force development.

† = significant at $p < 0.05$.

‡ = significant at $p < 0.01$.

However, one needs to be careful in applying the findings of the present investigation, because power output was measured for only 1 repetition. Weightlifting movements and their derivatives often are performed for multiple repetitions. When performing multiple repetitions of the hang power clean and other resistance-training exercises, it is expected that power output will progressively decrease as neuromuscular fatigue manifests itself (19). Therefore, investigating power output during multiple repetitions of the hang power clean and other resistance-training exercises would provide further information useful to athletes. In addition, athletes should consider the

use of a cluster set configuration. The cluster set configuration refers to performing a training set with 10–30-second rest intervals between repetitions, as opposed to the traditional set configuration, which comprises all repetitions that are performed in a continuous fashion without rest between repetitions (19, 43). Haff et al. (19) investigated the effect of different types of set configurations on barbell kinematics during performance of the clean pull. They suggested that the cluster set configuration maintains resistance-training performance during multiple repetitions, whereas the traditional set configuration gradually decreases resistance-training perfor-

TABLE 9. The comparisons of 1RM and peak and average power at different relative intensities during the hang power clean—7 strong vs. 8 weak subjects (Mean $\pm$ SD).*

Variable	7 strong (1RM $\geq$ 110 kg)	8 weak (1RM < 110 kg)	<i>p</i> value	Effect size
1RM (kg)	120.7 $\pm$ 15.9	95.0 $\pm$ 11.6	0.003	0.48
1RM (kg·BdM ⁻¹)	1.29 $\pm$ 0.17	1.13 $\pm$ 0.06	0.024	0.32
Peak power (W)				
At 30%	2,873.01 $\pm$ 684.06	3,114.00 $\pm$ 948.23	0.587	0.02
At 40%	3,587.75 $\pm$ 677.52	3,245.22 $\pm$ 936.88	0.438	0.04
At 50%	3,774.08 $\pm$ 579.23	3,571.40 $\pm$ 970.90	0.638	0.02
At 60%	3,983.08 $\pm$ 630.69	3,835.23 $\pm$ 820.96	0.705	0.01
At 70%	4,281.15 $\pm$ 634.84	3,868.52 $\pm$ 702.17	0.257	0.09
At 80%	4,070.70 $\pm$ 518.09	3,982.58 $\pm$ 906.49	0.825	0.00
At 90%	4,193.24 $\pm$ 514.92	3,633.86 $\pm$ 661.51	0.094	0.19
Average power (W)				
At 30%	1,578.57 $\pm$ 399.81	1,138.54 $\pm$ 444.22	0.066	0.24
At 40%	1,863.37 $\pm$ 339.46	1,443.65 $\pm$ 506.36	0.086	0.21
At 50%	2,040.50 $\pm$ 405.42	1,546.08 $\pm$ 515.65	0.062	0.24
At 60%	2,193.13 $\pm$ 370.28	1,651.50 $\pm$ 438.02	0.024	0.34
At 70%	2,303.34 $\pm$ 391.49	1,618.45 $\pm$ 389.62	0.005	0.47
At 80%	2,229.37 $\pm$ 357.82	1,623.32 $\pm$ 312.21	0.004	0.49
At 90%	2,262.81 $\pm$ 520.13	1,559.95 $\pm$ 289.54	0.006	0.46

* 1RM = 1 repetition maximum; BdM = body mass.

mance due to fatigue. Therefore, training at the optimal load, combined with the use of the cluster set configuration for multiple repetitions, may maximize and maintain power output, which may provide a better stimulus for maximal power development and athletic performance enhancement.

Another consideration for athletes and coaches is that when performing power training at the optimal load, the effort to produce force should be maximized (i.e., maximum rates of force development). Although 70% 1RM was found to maximize power output during the hang power clean in the present investigation, it is possible that athletes could produce submaximal effort and still complete the repetitions successfully with submaximal loads. In such a case, power output would be submaximal, which results in a submaximal stimulus for the development of muscular power. In fact, it has been suggested that training with the voluntary effort of producing force as fast as possible results in superior gains in power and explosive strength, as compared with training with an intentionally slower muscle action (14, 55). Therefore, athletes should perform the hang power clean as explosively as possible in an attempt to maximize power output during power training, regardless of the weights used. Additionally, because there is a lack of visual feedback of power output during resistance training, the development and use of any technology that enables immediate feedback of power output and is available at affordable cost may be desirable.

There was no significant correlation between power output during the hang power clean and vertical jump displacement, in spite of the importance of power suggested for activities such as jumping (39) and in spite of the mechanical specificity reported between these 2 activities (8). However, when converted to relative values, peak power per body weight at relatively high percentages of 1RM (>70%) did show strong to very strong correlations with vertical jump displacements, indicating that relative power production is more important for unloaded jumping than absolute power production is.

Power output during the hang power clean, especially

peak power, showed moderate to very strong correlations with peak power during the vertical jump. In addition, power output during the hang power clean showed strong to very strong correlations with peak force during vertical jump. Therefore, it seems that these two movements share some structural and functional neuromuscular foundation, a concept that also was supported by moderate to very strong correlations between peak force during the hang power clean and peak power and peak force during the vertical jump.

In addition to muscular power, force-time curve characteristics of muscle actions are considered important and have been investigated with respect to dynamic athletic performance (18, 34, 37, 41, 51, 52, 54). Traditionally, force-time curve analyses have been performed using isometric testing methods with conflicting results concerning the relationship to dynamic athletic performance (18, 34, 36, 51, 52). Although several investigators suggest that isometric force-time curve evaluation has limited application to assessing the dynamic capacity of the neuromuscular system and that the use of dynamic measurement of force-time curve is superior to isometric measurement (36, 38, 41, 52), little data exist in the scientific literature concerning the dynamic assessment of force-time curves, and further research is necessary.

In the present investigation, force-time curve characteristics such as peak force and PRFD were assessed during the stretch-shortening cycle (SSC) of a dynamic muscle action performed at different relative intensities. Peak force increased as a function of loads, agreeing with previous studies (37, 40, 45), whereas relative intensity had little effect on PRFD, which also agrees with the result of a previous investigation (41). Regarding the relationship with dynamic athletic performance, neither peak force nor PRFD was significantly correlated with vertical jump displacement, regardless of the load used. This finding is consistent with that of a study by Wilson et al. (52), who found no relationship between the SSC force-time curve tests and sprint performance. The lack of relationship of force-time curve variables during the hang power clean with SJ displacement in the present study could be

attributed to the different types of muscle action used (SSC muscle action for the hang power clean vs. concentric-only muscle action for the SJ). On the other hand, it was expected that the force-time curve variables generated during the hang power clean would be related significantly to CMJ displacement based on muscle-action type specificity (i.e., both were SSC movements). However, the force-time curve variables recorded during the hang power clean were not related to CMJ displacement. This may be due to the differences in magnitude and velocity of the countermovement between the hang power clean and the CMJ. Interestingly, Wilson et al. (52) demonstrated the superiority of concentric-only RFD tests to SSC tests, with respect to the relationship with dynamic SSC performance in spite of a lack of muscle-action type specificity. Therefore, one might need to consider the use of concentric-only force-time curve tests instead of SSC tests in order to evaluate dynamic neuromuscular function that is related to dynamic SSC performance. However, further research is necessary to validate the use of concentric tests.

As for the influence of 1RM on the optimal load for the highest power output, there exist 2 conflicting observations. Baker (3) reported that athletes with a higher 1RM tend to use lower percentages of 1RM to maximize power output during the bench press throw and the jump squat, as compared with athletes with a lower 1RM. On the other hand, Stone et al. (48) found the opposite trend, suggesting that there may be an upward shift in the optimal load for the highest power output as a result of an increased 1RM. Part of the reason for this difference is that Stone et al. (48) used a group of heterogeneously trained subjects who differed considerably in 1RM, whereas Baker (3) used a group of homogeneously trained subjects who also were experienced in power training method. Therefore, it could be speculated that an initial increase in maximum strength leads to an upward shift in the optimal load, whereas further training using power training method with lighter weights leads to a downward shift. In the present investigation, subjects with a higher 1RM achieved the highest peak power at a lower percentage of 1RM (70%) during the hang power clean than subjects with a lower 1RM (80%), whereas subjects with a higher 1RM achieved the highest average power at a higher percentage of 1RM (70%) than subjects with a lower 1RM (60%). Obviously, further research is warranted in order to conclude how 1RM influences the relative intensity at which the highest power output occurs.

PRACTICAL APPLICATIONS

When designing resistance-training programs, various factors need to be considered. Training load used is one factor that can affect the efficacy of the resistance-training program. The results of this study indicate that power output is maximized at submaximal loads, rather than at near-maximal loads, during the hang power clean. Training at a load that maximizes power output is recommended in order to improve maximal muscular power and dynamic athletic performance. This type of power training should be accompanied by a voluntary effort to produce force as fast as possible in an attempt to maximize power output. Furthermore, when power training is performed for multiple repetitions, the use of a cluster set configuration should be considered in order to maximize

and maintain power output and to gain a maximum training stimulus.

It should also be noted, however, that power training with loads that maximize power output is not the only option for athletes. Overall resistance-training programs should be periodized so that various factors such as strength, power, strength endurance, speed, and PRFD are addressed in a sequential manner. Therefore, in order to benefit from power training with optimal loads, such training should be incorporated into overall periodized training programs so that athletes can gain maximum training adaptations and improve athletic performance.

REFERENCES

1. BAKER, D. Improving vertical jump performance through general, special, and specific strength training: A brief review. *J. Strength Cond. Res.* 10:131-136. 1996.
2. BAKER, D. Acute and long-term power responses to power training: observations on the training of an elite power athlete. *Strength Cond. J.* 23:47-56. 2001.
3. BAKER, D. A series of studies on the training of high-intensity muscle power in rugby league football players. *J. Strength Cond. Res.* 15:198-209. 2001.
4. BAKER, D., S. NANCE, AND M. MOORE. The load that maximizes the average mechanical power output during explosive bench press throws in highly trained athletes. *J. Strength Cond. Res.* 15:20-24. 2001.
5. BAKER, D., S. NANCE, AND M. MOORE. The load that maximizes the average mechanical power output during jump squats in power-trained athletes. *J. Strength Cond. Res.* 15:92-97. 2001.
6. BEMBEN, M.G., J.L. CLASEY, AND B.H. MASSEY. The effect of the rate of muscle contraction on the force-time curve parameters of male and female subjects. *Res. Q. Exerc. Sport* 61:96-99. 1990.
7. BOSCO, C., P. LUHTANEN, AND P.V. KOMI. A simple method for measurement of mechanical power in jumping. *Eur. J. Appl. Physiol. Occup. Physiol.* 50:273-282. 1983.
8. BURKHARDT, E., AND J. GARHAMMER. Biomechanical comparison of hang cleans and vertical jumps. *J. Appl. Sport. Sci. Res.* 2:57. 1988.
9. COHEN, J. *Statistical Power Analysis for the Behavioral Sciences* (2nd ed.). Hillsdale, NJ: Lawrence Erlbaum, 1988.
10. CRONIN, J., P.J. MCNAIR, AND R.N. MARSHALL. Developing explosive power: A comparison of technique and training. *J. Sci. Med. Sport* 4:59-70. 2001.
11. DUCHATEAU, J., AND K. HAINAUT. Isometric or dynamic training: Differential effects on mechanical properties of a human muscle. *J. Appl. Physiol.* 56:296-301. 1984.
12. ENOKA, R.M. The pull in Olympic weightlifting. *Med. Sci. Sports* 11:131-137. 1979.
13. FAULKNER, J.A., D.R. CLAFLIN, AND K.K. MCCULLY. Power output of fast and slow fibers from human skeletal muscles. In: *Human Muscle Power*. N.L. Jones, N.M. McCartney, and A.J. McComas, eds. Champaign, IL: Human Kinetics, 1986. pp. 81-94.
14. FIELDING, R.A., N.K. LEBRASSEUR, A. CUOCO, J. BEAN, K. MIZER, AND M.A. FIATARONE SINGH. High-velocity resistance training increases skeletal muscle peak power in older women. *J. Am. Geriatr. Soc.* 50:655-662. 2002.
15. GARHAMMER, J. Weight lifting and training. In: *Biomechanics of Sport*. C.L. Vaughan, ed. Boca Raton, FL: CRC Press, 1989. pp. 169-211.
16. GARHAMMER, J. A review of power output studies of Olympic and powerlifting: Methodology, performance prediction, and evaluation tests. *J. Strength Cond. Res.* 7:76-89. 1993.
17. GARHAMMER, J., AND R. GREGOR. Propulsion forces as a function of intensity for weightlifting and vertical jumping. *J. Appl. Sport. Sci. Res.* 6:129-134. 1992.

18. HAFF, G.G., M. STONE, H.S. O'BRYANT, E. HARMAN, C. DINAN, R. JOHNSON, AND K.-H. HAN. Force-time dependent characteristics of dynamic and isometric muscle actions. *J. Strength Cond. Res.* 11:269–272. 1997.
19. HAFF, G.G., A. WHITLEY, L.B. MCCOY, H.S. O'BRYANT, J.L. KILGORE, E.E. HAFF, K. PIERCE, AND M.H. STONE. Effects of different set configurations on barbell velocity and displacement during a clean pull. *J. Strength Cond. Res.* 17:95–103. 2003.
20. HAFF, G.G., A. WHITLEY, AND J.A. POTTEIGER. A brief review: Explosive exercises and sports performance. *Strength Cond. J.* 23:13–20. 2001.
21. HÄKKINEN, K., H. KAUKANEN, AND P.V. KOMI. Biomechanical changes in the olympic weightlifting technique of the snatch and clean & jerk from submaximal to maximal loads. *Scand. J. Sports Sci.* 6:57–66. 1984.
22. HARMAN, E.A., M.T. ROSENSTEIN, P.N. FRYKMAN, AND R.M. ROSENSTEIN. The effects of arms and countermovement on vertical jumping. *Med. Sci. Sports Exerc.* 22:825–833. 1990.
23. HARMAN, E.A., M.T. ROSENSTEIN, P.N. FRYKMAN, R.M. ROSENSTEIN, AND W.J. KRAEMER. Estimation of human power output from vertical jump. *J. Appl. Sport Sci. Res.* 5:116–120. 1991.
24. HOLM, S. A simple sequentially rejective multiple test procedure. *Scand. J. Stat.* 6:65–70. 1979.
25. HOPKINS, W.G. A scale of magnitude for effect statistics. A new view of statistics. Last updated 2003. Available at: <http://sportsci.org/resource/stats/>. Accessed May 4, 2005.
26. IZQUIERDO, M., K. HÄKKINEN, J.J. GONZALEZ-BADILLO, J. IBANEZ, AND E.M. GOROSTIAGA. Effects of long-term training specificity on maximal strength and power of the upper and lower extremities in athletes from different sports. *Eur. J. Appl. Physiol.* 87:264–271. 2002.
27. IZQUIERDO, M., J. IBANEZ, E. GOROSTIAGA, M. GARRUES, A. ZUNIGA, A. ANTON, J.L. LARRION, AND K. HÄKKINEN. Maximal strength and power characteristics in isometric and dynamic actions of the upper and lower extremities in middle-aged and older men. *Acta Physiol. Scand.* 167:57–68. 1999.
28. JACKSON, A.S., AND M.L. POLLOCK. Generalized equations for predicting body density of men. *Br. J. Nutr.* 40:497–504. 1978.
29. KANEKO, M., T. FUCHIMOTO, H. TOJI, AND K. SUEI. Training effects of different loads on the force-velocity relationship and mechanical power output in human muscle. *Scand. J. Sports Sci.* 5:50–55. 1983.
30. KAWAMORI, N., AND G.G. HAFF. The optimal training load for the development of muscular power. *J. Strength Cond. Res.* 18: 674–684. 2004.
31. KIRKSEY, B., M.H. STONE, B.J. WARREN, R.L. JOHNSON, M. STONE, G.G. HAFF, F.E. WILLIAMS, AND C. PROULX. The effects of 6 weeks of creatine monohydrate supplementation on performance measures and body composition in collegiate track and field athletes. *J. Strength Cond. Res.* 13:148–156. 1999.
32. MACALUSO, A., AND G. DE VITO. Comparison between young and older women in explosive power output and its determinants during a single leg-press action after optimisation of load. *Eur. J. Appl. Physiol.* 90:458–463. 2003.
33. MARCORA, S., AND M.K. MILLER. The effect of knee angle on the external validity of isometric measures of lower body neuromuscular function. *J. Sports Sci.* 18:313–319. 2000.
34. MERO, A., P. LUHTANEN, J.T. VIITASALO, AND P.V. KOMI. Relationships between the maximal running velocity, muscle fiber characteristics, force production and force relaxation of sprinters. *Scand. J. Sports Sci.* 3:16–22. 1981.
35. MIRKOV, D.M., A. NEDELJKOVIC, S. MILANOVIC, AND S. JARIC. Muscle strength testing: evaluation of tests of explosive force production. *Eur. J. Appl. Physiol.* 91:147–154. 2004.
36. MURPHY, A.J., AND G.J. WILSON. Poor correlations between isometric tests and dynamic performance: Relationship to muscle activation. *Eur. J. Appl. Physiol. Occup. Physiol.* 73:353–357. 1996.
37. MURPHY, A.J., G.J. WILSON, AND J.F. PRYOR. Use of the isoinertial force mass relationship in the prediction of dynamic human performance. *Eur. J. Appl. Physiol. Occup. Physiol.* 69: 250–257. 1994.
38. MURPHY, A.J., G.J. WILSON, J.F. PRYOR, AND R.U. NEWTON. Isometric assessment of muscular function: The effect of joint angle. *J. Appl. Biomech.* 1:205–215. 1995.
39. NEWTON, R.U., AND W.J. KRAEMER. Developing explosive muscular power: implications for a mixed methods training strategy. *Strength Cond. J.* 16:20–31. 1994.
40. NEWTON, R.U., A.J. MURPHY, B.J. HUMPHRIES, G.J. WILSON, W.J. KRAEMER, AND K. HÄKKINEN. Influence of load and stretch shortening cycle on the kinematics, kinetics and muscle activation that occurs during explosive upper-body movements. *Eur. J. Appl. Physiol. Occup. Physiol.* 75:333–342. 1997.
41. PRYOR, J.F., G.J. WILSON, AND A.J. MURPHY. The effectiveness of eccentric, concentric and isometric rate of force development tests. *J. Hum. Mov. Stud.* 27:153–172. 1994.
42. SIEGEL, J.A., R.M. GILDERS, R.S. STARON, AND F.C. HAGERMAN. Human muscle power output during upper- and lower-body exercises. *J. Strength Cond. Res.* 16:173–178. 2002.
43. SIFF, M.C. *Supertraining*. Denver: Supertraining Institute, 2004.
44. SIRI, W.E. Body composition from fluid spaces and density: Analysis of methods. In: *Techniques for Measuring Body Composition*. J. Brozek and A. Henschel, eds. Washington, DC: National Academy of Sciences, 1961. pp. 223–244.
45. SLEIVERT, G., AND M. TAINGAHUE. The relationship between maximal jump-squat power and sprint acceleration in athletes. *Eur. J. Appl. Physiol.* 91:46–52. 2004.
46. STONE, M.H. Position statement and literature review: Explosive exercises and training. *Natl. Strength Cond. Assoc. J.* 15: 9–15. 1993.
47. STONE, M.H., AND H.S. O'BRYANT. *Weight Training: A Scientific Approach*. Minneapolis: Burgess International, 1987.
48. STONE, M.H., H.S. O'BRYANT, L. MCCOY, R. COGLIANESE, M. LEHMKUHL, AND B. SCHILLING. Power and maximum strength relationships during performance of dynamic and static weighted jumps. *J. Strength Cond. Res.* 17:140–147. 2003.
49. STONE, M.H., S. PLISK, AND D. COLLINS. Training principles: Evaluation of modes and methods of resistance training—A coaching perspective. *Sports. Biomech.* 1:79–103. 2002.
50. THOMAS, M., M.A. FIATARONE, AND R.A. FIELDING. Leg power in young women: relationship to body composition, strength, and function. *Med. Sci. Sports Exerc.* 28:1321–1326. 1996.
51. VIITASALO, J.T., AND O. AURA. Seasonal fluctuations of force production in high jumpers. *Can. J. Appl. Sport Sci.* 9:209–213. 1984.
52. WILSON, G.J., A.D. LYTTL, K.J. OSTROWSKI, AND A.J. MURPHY. Assessing dynamic performance: A comparison of rate of force development tests. *J. Strength Cond. Res.* 9:176–181. 1995.
53. WILSON, G.J., R.U. NEWTON, A.J. MURPHY, AND B.J. HUMPHRIES. The optimal training load for the development of dynamic athletic performance. *Med. Sci. Sports Exerc.* 25:1279–1286. 1993.
54. YOUNG, W., B. MCLEAN, AND J. ARDAGNA. Relationship between strength qualities and sprinting performance. *J. Sports Med. Phys. Fitness* 35:13–19. 1995.
55. YOUNG, W.B., AND G.B. BILBY. The effect of voluntary effort to influence speed of contraction on strength, muscular power, and hypertrophy development. *J. Strength Cond. Res.* 7:172–178. 1993.

Acknowledgments

We would like to thank all the subjects who participated in this study and Head Football Coach Bill Maskill for their support for this project.

Address correspondence to G. Gregory Haff, WVU School of Medicine, Division of Exercise Physiology, PO Box 9227, Morgantown, WV 26506.